APM1220: Applied Multivariate Data Analysis
FA 1

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Aug 27, 2026

Problem 2.2

Given the matrices $A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}$, and $C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}$, perform the indicated operations:

(a) $5A$

$$5A = 5\begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}$$

(b) BA

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}\begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 4(-1)-3(4) & 4(3)-3(2) \\ 1(-1)-2(4) & 1(3)-2(2) \\ -2(-1)+0(4) & -2(3)+0(2) \end{bmatrix} = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

(c) $A'B'$ Using the property $A'B' = (BA)'$, we transpose the result from (b):

$$A'B' = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}$$

(d) $C'B$

$$C'B = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix}\begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 5(4)-4(1)+2(-2) & 5(-3)-4(-2)+2(0) \end{bmatrix} = \begin{bmatrix} 12 & -7 \end{bmatrix}$$

(e) Is AB defined? No. Matrix A has dimensions 2×2 and matrix B has dimensions 3×2. The number of columns in A (2) does not equal the number of rows in B (3).

Problem 2.4

(a) Prove $(A')^{-1} = (A^{-1})'$

By definition of an inverse, $AA^{-1} = I$. Taking the transpose of both sides gives $(AA^{-1})' = I'$. Since $I' = I$ and the transpose of a product is the product of the transposes in reverse order, we have $(A^{-1})'A' = I$. Multiplying both sides on the right by $(A')^{-1}$, we get $(A^{-1})' = (A')^{-1}$.

(b) Prove $(AB)^{-1} = B^{-1}A^{-1}$ We evaluate the product of AB and its proposed inverse: $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B$. Since $A^{-1}A = I$, this simplifies to $B^{-1}IB = B^{-1}B = I$. Since multiplying AB by $B^{-1}A^{-1}$ yields the identity matrix, $(AB)^{-1} = B^{-1}A^{-1}$.

Problem 2.8

Given $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$. To find the eigenvalues, solve the characteristic equation $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix} = (1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6 = (\lambda+3)(\lambda-2) = 0$$

Thus, the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -3$.

For $\lambda_1 = 2$:

$$(A - 2I)x = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -x_1 + 2x_2 = 0 \implies x_1 = 2x_2$$

The eigenvector is $v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Normalizing gives $e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

For $\lambda_2 = -3$:

$$(A + 3I)x = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 2x_1 + x_2 = 0 \implies x_2 = -2x_1$$

The eigenvector is $v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Normalizing gives $e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

The spectral decomposition is $A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2'$.

Problem 2.11

By Definition 2A.24, expanding the determinant along the first row of the diagonal matrix A, we get $|A| = a_{11}|A_{11}| + 0 + \cdots + 0$, where A_{11} is the submatrix obtained by deleting the first row and first column. Since A_{11} is also a diagonal matrix, we can repeat this process: $|A_{11}| = a_{22}|A_{22}|$. Continuing this process for all p dimensions yields the product of the diagonal elements: $|A| = a_{11}a_{22}\dots a_{pp}$.

Problem 2.24

Given $\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(a) Σ^{-1} For a diagonal matrix, the inverse is the reciprocal of the diagonal elements:

$$\Sigma^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Eigenvalues and eigenvectors of Σ Since Σ is diagonal, its eigenvalues are the diagonal entries: $\lambda_1 = 4, \lambda_2 = 9, \lambda_3 = 1$. The normalized eigenvectors are the standard basis vectors: $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

(c) Eigenvalues and eigenvectors of Σ^{-1} The eigenvalues of the inverse matrix are the reciprocals of the original eigenvalues: $\lambda_1 = 1/4, \lambda_2 = 1/9, \lambda_3 = 1$. The eigenvectors remain the same as in (b).

Problem 2.30

Partitioning X into $X^{(1)} = [X_1, X_2]'$ and $X^{(2)} = [X_3, X_4]'$. We partition μ_X into $\mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. We partition Σ_X into $\Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, $\Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$, and $\Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$.

- **(a)** $E(X^{(1)}) = \mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$
- **(b)** $E(AX^{(1)}) = A\mu^{(1)} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 10$
- **(c)** $Cov(X^{(1)}) = \Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$
- **(d)** $Cov(AX^{(1)}) = A\Sigma_{11}A' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 7$
- **(e)** $E(X^{(2)}) = \mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
- **(f)** $E(BX^{(2)}) = B\mu^{(2)} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

- **(g)** $Cov(X^{(2)}) = \Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$

- **(h)** $Cov(BX^{(2)}) = B\Sigma_{22}B' = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}$

- **(i)** $Cov(X^{(1)}, X^{(2)}) = \Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$

- **(j)** $Cov(AX^{(1)}, BX^{(2)}) = A\Sigma_{12}B' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix}$

Problem 2.41

Given $\mu_X = \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix}$ and $\Sigma_X = 3I_{4\times4}$.

(a) Find $E(AX)$

$$E(AX) = A\mu_X = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}$$

(b) Find $Cov(AX)$

$$Cov(AX) = A\Sigma_X A' = A(3I)A' = 3AA'$$

$$3 \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix} = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

(c) Which pairs of linear combinations have zero covariances? Since the resulting covariance matrix in (b) is diagonal, **all pairs** of linear combinations have zero covariance.